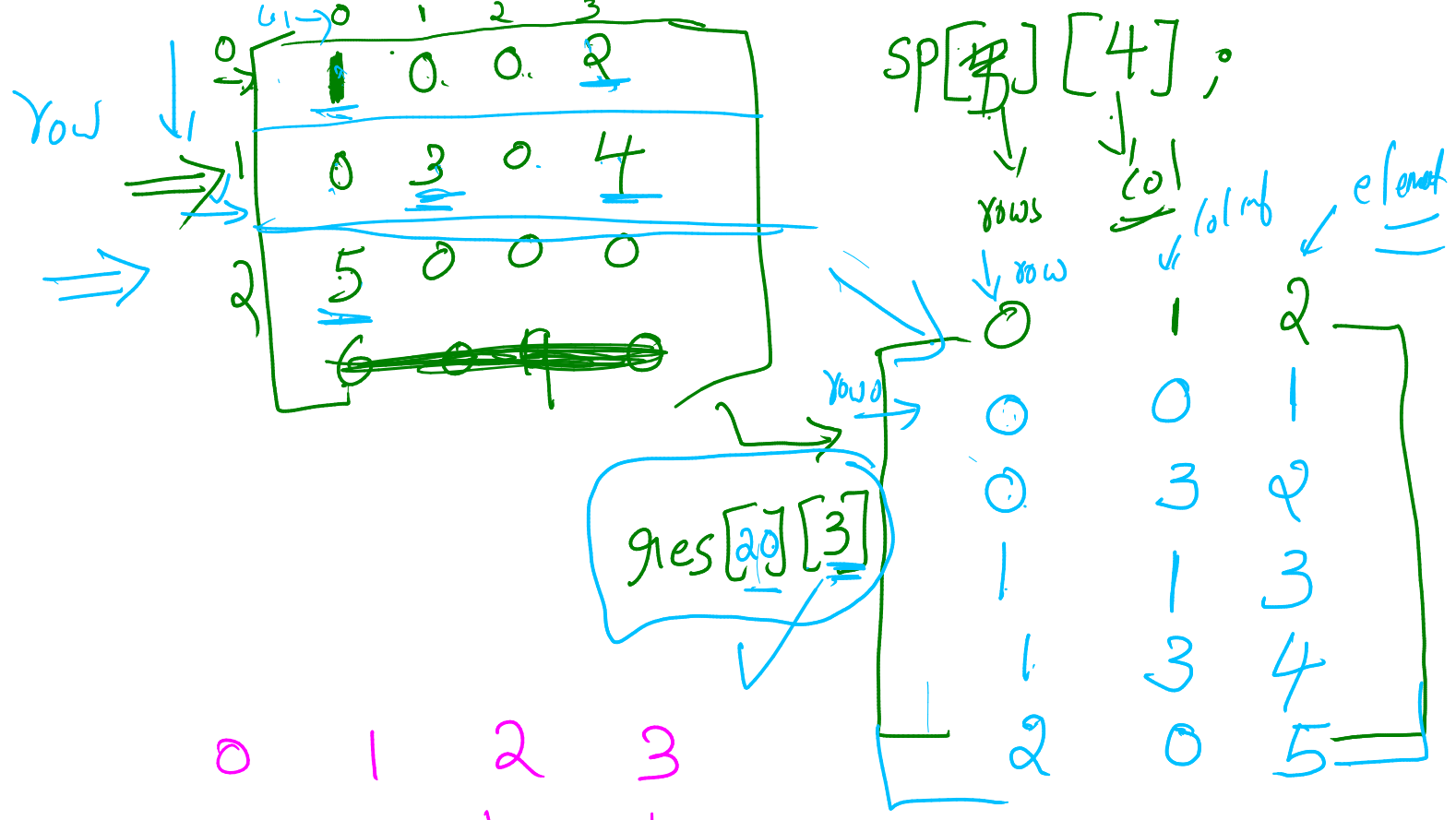
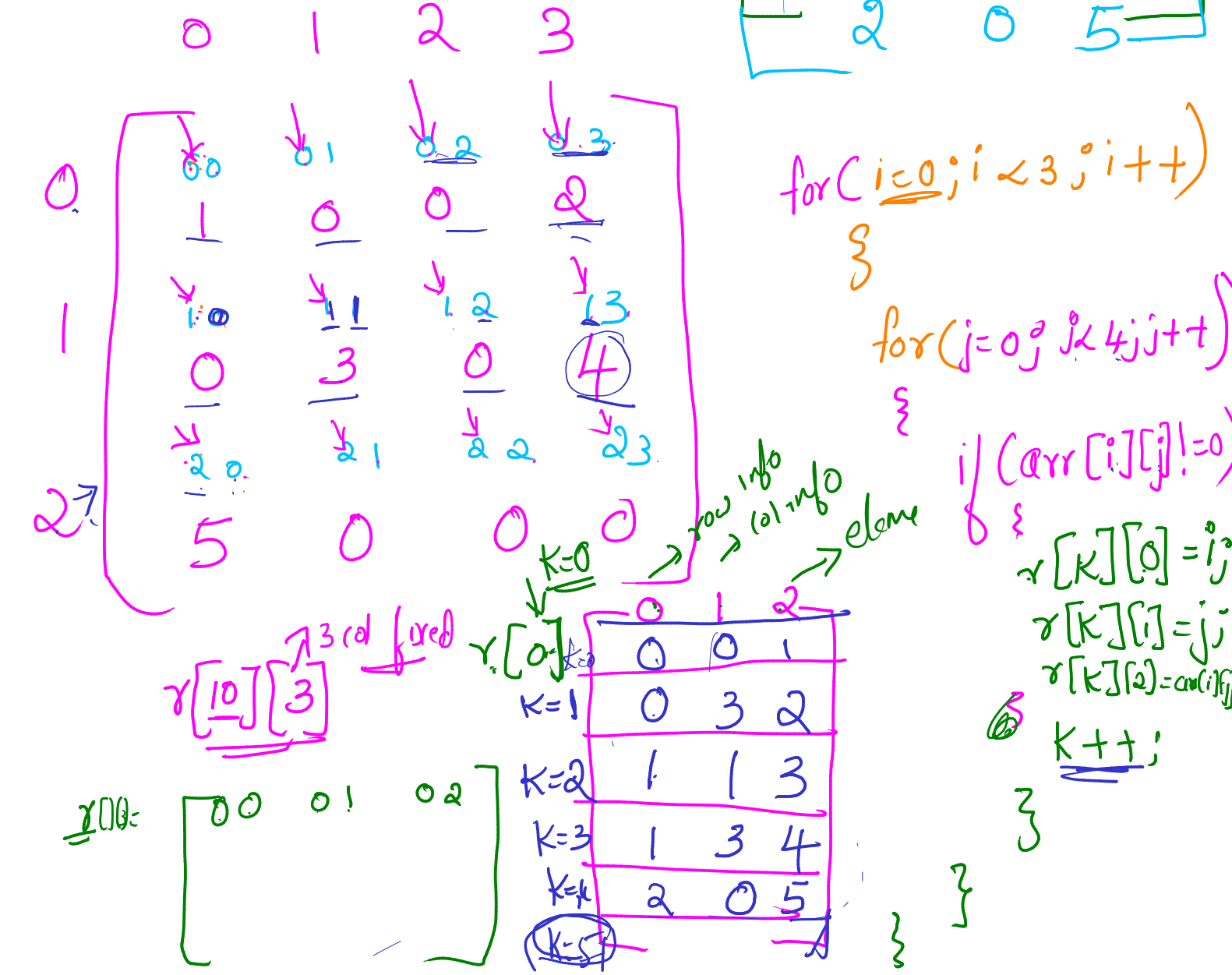

row
sp[ ][4];
rows
col
row
col info
element
res[20][3]
row


for(i=0; i<3; i++)
{
for(j=0; j<4; j++)
{
if(arr[i][j]!=0)
{
r[k][0]=i;
r[k][1]=j;
r[k][2]=arr[i][j];
k++;
}
}
}
r[10][3]
3 col fixed
r[0]
k=0
row info
col info
elem
r[][]=

```
for(int i=0; i<k; i++)        k=5
{
    for(int j=0; j<3; j++)
    {
        printf("%d", r[i][j]);
    }
    printf("\n");
}
```

row 0 1 2 3 4

~~4~~ ~~3~~ com ~~k=5~~

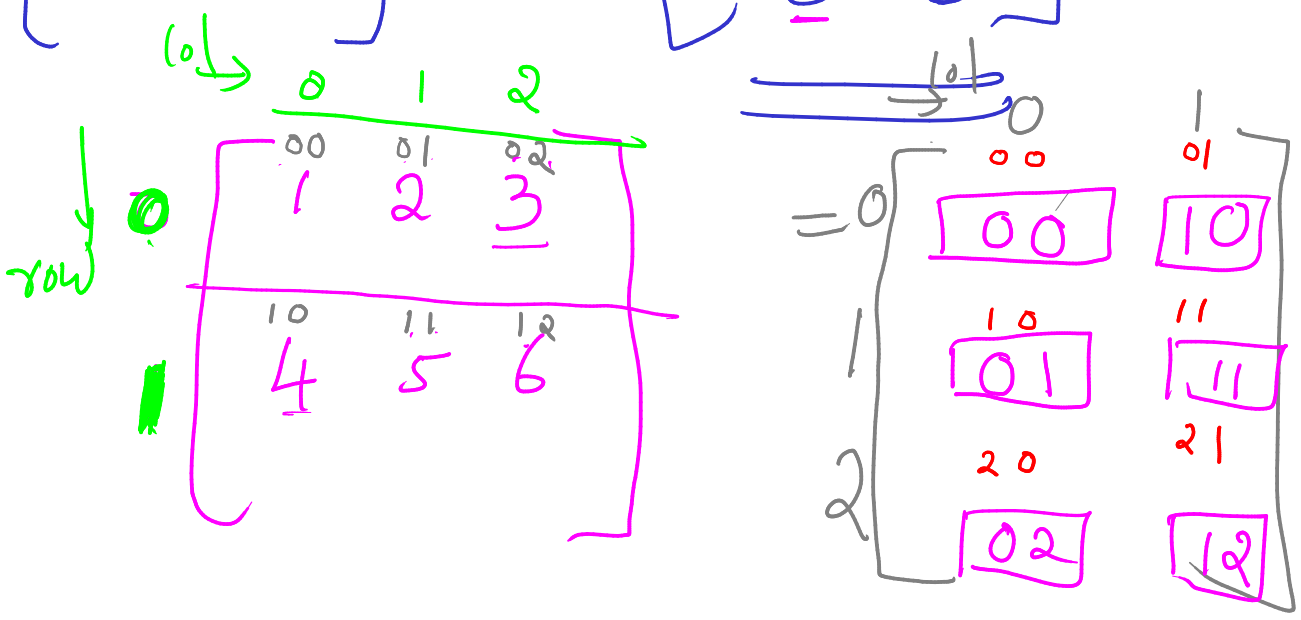
i=0 j=0

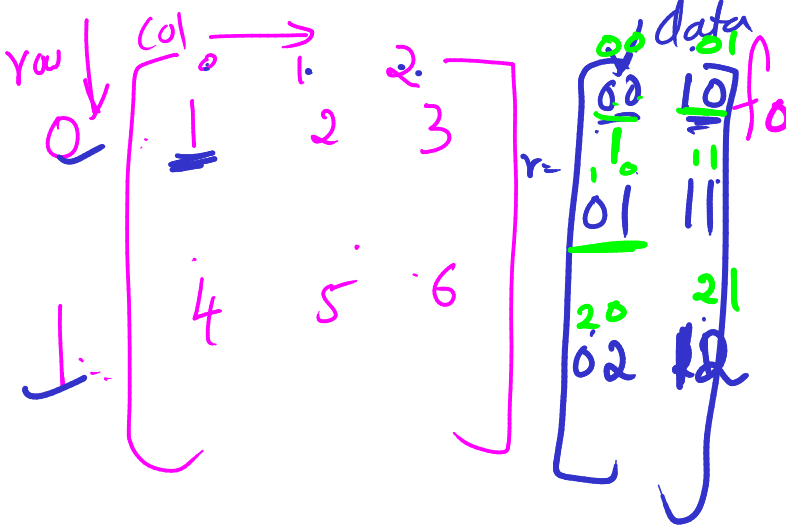
r[0][0] r[0][1] r[0][2]

r[1][0] r[1][1] r[1][2]

Transpose of a matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$






```
for(i=0; i<2; i++)
{
    for(j=0; j<3; j++)
    {
        r[j][i] = arr[i][j];
    }
}
```

i=0; i<2; j
j=0; j<3
r[0][0] = arr[0][0]

j=1, 1<3
= arr[i][j]
r[j][i] = arr[0][1]
r[1][0] = arr[0][1]
r[1][0] = 2

j=2, 2<3
i arr[0][j]
r[2][0] = arr[0][2]

j=3, 3<3 X

r	col 0	col 1
i=0 → 0	00	01
	1	4
1	10	11
	2	5
2	20	21
	3	6

i=1, 1<2 → j=0, 0<3
r[j][i] = arr[i][j]
r[0][1] = arr[1][0]

j=1
r[j][i] = arr[i][j]
r[1][1] = arr[1][1]

j=2
r[j][i] = arr[i][j]
r[2][1] = arr[1][2]